An Explainable Data-Driven Platform for Optimal Wind Farm Siting

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1 Introduction

Wind farm site selection is a crucial yet highly complex challenge. Optimal siting involves much more than simply identifying windy locations; it requires balancing multiple, often competing factors including, but not limited to, proximity to the electricity grid, consistency of wind resources, land use constraints, environmental concerns, community acceptance, and economic viability [1–3]. The goal is to select locations that will not only generate the most reliable energy over time but also integrate well into the broader power system and surrounding landscape. However, this decision-making process is often not systematically optimized.

This issue has become even more pressing in the context of growing climate change awareness and the global transition toward sustainable energy systems. In Australia, policy commitments such as the Victorian Renewable Energy Targets, legislated in the *Renewable Energy Act (2017)* [4], reinforce this shift. These targets include:

- 40% of Victoria's electricity generation to be renewable by 2025
- 65% by 2030
- 95% by 2035

Meeting such goals will demand not only increased renewable energy capacity, but also smarter, more coordinated deployment strategies. This underscores the need for data-driven and holistic planning tools that can guide decision-making.

Currently, wind farm development in Australia is largely driven by a bottom-up planning approach. While site selection occurs within a defined legal framework and requires approval from energy market regulators, final decisions are often shaped by the interests of local stakeholders, like landowners, and by wind resource assessments conducted over relatively short timescales. Given that wind farms typically operate for 20 to 30 years, this approach can result in long-term inefficiencies [5]. Studies have shown that such a fragmented planning process can lead to suboptimal outcomes in terms of both total energy generation and supply stability. In contrast, a top-down planning model, where wind farm locations are determined with long-term demand patterns in mind, has demonstrated the potential for significantly greater efficiency and reliability [6].

1.1 Aim

This literature review aims to explore existing methods for wind farm site selection, both globally and within Australia. It will examine traditional approaches as well as recent techniques that incorporate artificial intelligence, with a particular focus on how explainable these methods are, both in terms of technical transparency (e.g., explainable AI) and in their ability to clearly justify decisions to non-technical audiences.

Findings from the review will inform the design of a user-friendly, explainable decision-support platform intended for a diverse range of users. These can include both technical actors such as infrastructure planners, power operators (e.g., AEMO, VicGrid) and decision makers, as well as ordinary citizens and local communities. In this context, clear and interactive representations of site selection criteria can help build trust, reduce misunderstandings, and support more inclusive planning. This is particularly important because the question of where to produce renewable electricity is often controversial, and the acceptance of new energy infrastructures, especially by the communities living near proposed sites, is critical, and it can lead to tension between stakeholders with different priorities or levels of technical understanding. [2, 7–9].

Figure 1: Recent protests against wind farms in Australia. **(Center)** A protest in Wollongong against plans for an offshore wind farm, May 2025 (Photo: Anna Warr, The Advocate). **(Right)** A rally on the front lawns of Parliament House in Canberra, February 24 (Photo: Mike Bowers, The Guardian). **(Left)** A sign photographed by the author in Warrnambool, March 2025

1.2 Document structure

Following the introduction, this report presents a structured literature review divided into several sections. The review explores the types of commonly used input data and critically examines traditional and AI based methods for wind farm site selection, with a particular focus on how explainable the methods are. It includes comparisons between approaches used across different countries, as well as a section dedicated to Australian visual and interactive tools, mainly focusing on mapping systems. These tools, coming from different domains than wind energy planning, offer inspiration and practical features that will help shape the design of the proposed platform. The State of the Art Summary then identifies gaps in the existing literature, particularly the lack of explainable and accessible visual tools that allow the creation of ”what-if” scenarios. The Research Project Plan outlines a practical approach to addressing these gaps, including a section on the technical tools to be used, such as programming languages and frameworks. Finally, the Conclusion section highlights the key findings and provides a concise summary of the report.

2 Literature review

2.1 Data Inputs for Wind Farm Site Suitability Modeling

Before delving into the various methods used for wind farm site selection, it is important to first understand the types of data commonly employed in these studies. The selection of suitable wind farm sites is inherently data-driven, relying on a wide range of datasets to evaluate the feasibility, efficiency, and sustainability of potential locations. These datasets, which span technical, environmental, and social factors, are fundamental in determining the most appropriate sites for wind farm development. After reviewing numerous studies it becomes evident that the types of data used are similar, regardless of geographical location or the specific approach employed. The most common siting factors identified across these studies are summarized in the table below:

Siting Factor
Wind Speed
Distance to Protected or Wildlife Areas
Slope
Distance to Urban Centers
Distance to Roads
Distance to Transmission Lines or Substations
Life Cycle Costs
Distance to Airports
Distance to Water Bodies
Elevation
Distance to Animal Habitats or Migration Zones
Wind Power Density
Distance to Agricultural Areas
Flood Risk
Landslide Risk
Distance to Historic Places
Distance to Railroads
Distance to Military Zones

Table 1: Common siting factors used in (onshore) wind farm site selection

However, even with these shared factors, a significant challenge persists: the lack of standardization in how data is represented and utilized. As highlighted in [10], a consistent and standardized approach to data representation is absent across the studies. This variability in data handling can create discrepancies in the results and undermines the reliability of findings across different studies, especially if applied to the same region of interest. The wind speed, for example, may be categorized either as an economic factor [11, 12], or as a techincal factor [13, 14]; as a constraint [15] (setting a minimum threshold) or as an evaluation criterion [16] (taken into account in order to maximize potential energy output). Furthermore, the choice of data sources often remains unclear, with many studies failing to specify where their datasets were obtained [1, 13, 17–20], which hinders transparency and replicability in site selection research.

2.2 Methods

The methods used for wind farm site selection (WiFSS) have grown more sophisticated over the past decade, driven by the growing demand for renewable energy. In general, these methods can be grouped into two main categories:

- Multi Criteria Decision Making **(MCDM)** methods
- **Non-MCDM** methods

Each group including a range of different techniques, differing in how they handle and process data.

2.2.1 Multi Criteria Decision Making (MCDM) methods

MCDM has been an active area of research since the 1970s [21] and its methods are applied across a wide range of research fields, including mathematics, energy systems, computer science, and economics [22]. MCDM methods addresses decision problems involving multiple, often conflicting criteria, typically measured in incomparable units. MCDM methods are designed to support the evaluation and selection of alternatives in complex scenarios where trade-offs are inevitable [23], such as in wind farm site selection. Determining importance coefficients (weights) for each factor of the problem is the central challenge in MCDM methods.

Analytic Hierarchy Process (AHP) In the context of wind farm site selection, the Analytic Hierarchy Process is probably the most extensively applied MCDM method, with numerous studies employing it across different regions in England [24], Saudi Arabia [25], Ecuador [26], India [27], Spain [28], among others. AHP is a rank-order weighting method, which means that each factor is considered to have a different level of importance when determining the best location for a wind farm. This method is composed of two key stages: pairwise comparisons and weighted linear combination (WLC). The pairwise comparison stage is needed to assign relative importance weights to the factors of the problem (e.g. distance from electrical grid, averaged wind speed). While WLC is used in the final step to evaluate and rank each potential wind farm location based on the calculated weights [29]. For instance, one wind farm location may have a certain distance from a national park, a specific distance from the electrical grid, and a particular average wind speed. Another location, in turn, might have different values for these same factors. Experts, using their knowledge and judgment, compare the criteria by performing pairwise scenario comparisons. Their opinions are then used to generate a comparison matrix, which is subsequently used to establish the relative weights. These weights determine which factors are more or less important in the context of the overall goal of identifying the best location. The main advantage of this method lies in its ability to break down a large problem into smaller, more manageable subproblems (such as comparing pairs of scenarios at different hierarchical levels) [30]. On the other hand, a major challenge is the reliance on expert opinions, which can be difficult to obtain and select. Moreover, these opinions can be conflicting and uncertain, as they are based on human judgment, which inherently involves subjectivity [1]. To overcome this issue, such methods also embed the calculation of a consistency index, which serves as a proxy for the logical coherence of judgments and helps ensure that the decision making process remains as reliable as possible [23].

Fuzzy Analytic Hierarchy Process (FAHP) Despite the use of a consistency index, the traditional AHP methods still faces limitations in handling the ambiguity and vagueness inherent in human judgment. Experts evaluations are often imprecise or linguistically expressed (e.g., "slightly more important" or "being far from the electrical grid is much less favorable than being far from a national park"), which makes it challenging to make precise judgments in pairwise comparisons. To better capture this uncertainty, the Fuzzy Analytic Hierarchy Process (FAHP) extends the conventional AHP by incorporating fuzzy set theory [31]. In a notable application of these principles, a study focusing on on-shore wind farm site selection in South Korea [1] employed the FAHP method to determine factors weights more robustly. Similar approaches have also been used in other countries, such as Greece [32] and Nigeria [33].

Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) TOPSIS is a popular MCDM technique, originally introduced over four decades ago [34], and still remains (together with its variations) one of the most widely applied methods in multi-criteria analysis. The core idea is intuitive: the optimal alternative should be closest to the ideal solution, representing the best possible performance across all criteria, and farthest from the negative ideal solution, which represents the worst case scenario. By calculating the relative distances to these two reference points, TOPSIS provides a simple ranking of the alternatives. Its appeal lies in its simplicity and compatibility with a variety of decision making contexts. For example, in the context of wind farm site selection, the ideal location might correspond to high wind speed, a short distance to the grid, and low environmental impact. These criteria can be represented as a vector. Each site in the region of interest is also represented as a vector of values and then compared to the ideal and negative-ideal vectors. TOPSIS calculates the geometric distance of each site to both extremes and assigns a relative closeness score. The site with the highest score is considered the most suitable. TOPSIS allows for the inclusion of weights to reflect the relative importance of factors, adjusting the distance computations to the ideal and negative-ideal solutions. This is why techniques like AHP and TOPSIS are often used together. For example, in a study conducted in Turkey [35], a FAHP method was applied to determine the weights of the factors. These weights were then used in the TOPSIS framework to rank the alternatives. A similar idea was used in Eastern Macedonia and Thrace [36]. However, while TOPSIS offers an intuitive solution to the problem, it is not without limitations. A key concern arises during the data normalization phase, where vector representations are constructed. Since TOPSIS is highly sensitive to the way data is normalized, selecting an appropriate normalization method is critical. Poor choices at this stage can lead to distorted distances and, ultimately, unreliable rankings. This is particularly problematic when dealing with factors that have different units or highly skewed distributions. Recent studies have highlighted this issue [37, 38]. Additionally, TOPSIS during the distances caluclation assumes that criteria are statistically independent [39], overlooking potential interactions that may exist between them. Finally, the method relies on human input to define the ideal and negative ideal solutions, which introduces subjectivity into the process. As with Fuzzy AHP, fuzzy extensions of TOPSIS have also been developed to address this uncertainty [40].

The methods discussed so far, regardless of their specific type, commonly utilize Geographic Information Systems (GIS) to visualize the results. GIS refers to a system designed to capture, store, manipulate, analyze, manage, and present geographical data. In the context of wind farm site selection, GIS tools are essential for integrating various spatial datasets. These systems allow for the visualization of complex, multi dimensional data in a geographical context, making it easier to identify locations that are more suitable or less suitable. Each of the methods typically produces a ranking or a score for each location. This ranking or score can then be used as a form of legend on the GIS map. Several examples are provided in Figure 2.

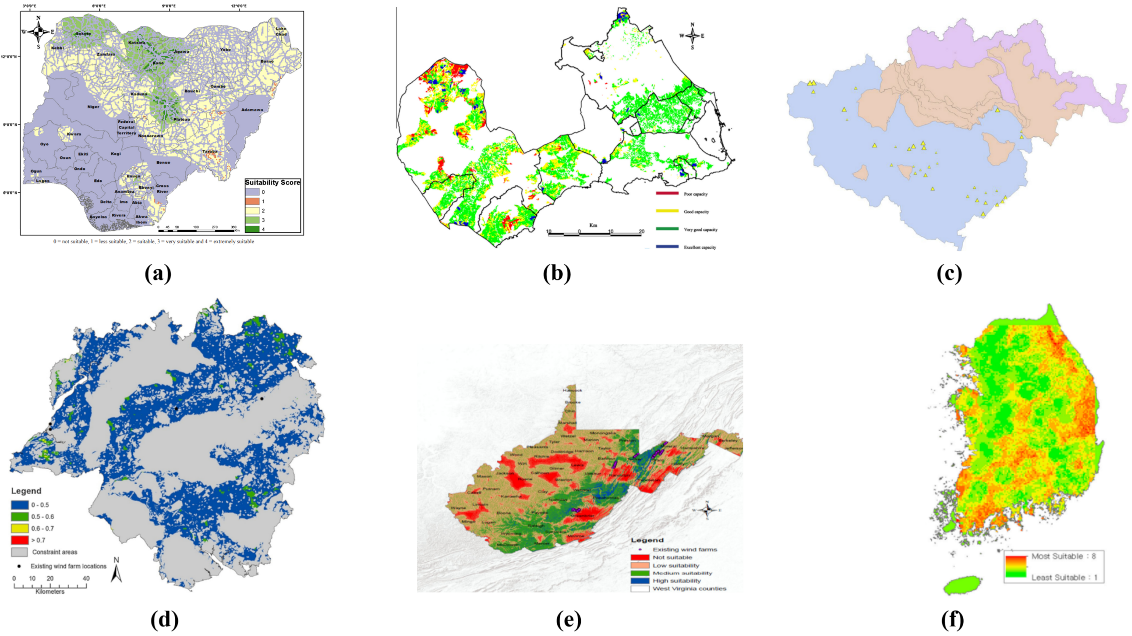
BBWM-based MCDM weighting results Includi immagine da: A spatio-temporal decision-making model for solar, wind, and hybrid systems – A case study of Saudi Arabia

Parte di AI Cita questo per la configurazione : Optimization of the Wind Turbine Layout and Transmission System Planning for a Large-Scale Offshore Wind Farm by AI Technology

Una parte su gunn

BBWM-based MCDM weighting results

Gap Mostra foto della mappa fino al 2022, e poi un solo studio dopo il 2022 Non spiegabili, static maps, not what if scenarions easily done (it requeires new computation) No python integration for future models (climate change)



(a) (b) (c) (d) (e) (f)

Figure 2: GIS-based visualizations showing the spatial distribution of site suitability for wind farm development. Regardless of the specific method used, each map represents suitability scores or rankings derived from multiple criteria.
(a) Nigeria, FAHP method [33] **(b)** Murcia region, Spain, FAHP method [28] **(c)** East Macedonia, Greece, AHP+TOPSIS method. Bigger triangles represent higher suitability [36] **(d)** South Central England, AHP method [16] **(e)** West Virginia, USA, AHP method [13] **(f)** South Korea, FAHP method [1]

3 Australian visual interfaces

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